

Timestamp Bug: Root Cause & Fix

SLT WhatsApp Chatbot Project — Debugging Session Summary — August 26, 2026

1. The Problem

Messages from the SLT WhatsApp API were displaying the wrong time in the chat UI. A message actually sent at **12:23** was showing as **17:53** on screen — consistently 5 hours 30 minutes ahead of the correct local time.

2. Root Cause

The bug was located in `extractIsoTimestamp()` inside `app/Services/ChatConversationService.php`. Timestamps returned by the SLT API (e.g. `"2026-08-26 12:23:16"`) do not include a timezone offset. The existing code assumed that any timestamp without an offset was in UTC, then converted it to Asia/Colombo (+5:30). Since the raw timestamp was already local Sri Lanka time, this added an extra +5:30 on top of a value that needed no adjustment — a double shift.

3. How It Was Confirmed

Rather than guessing, a temporary debug log line was added to print the raw timestamp on every incoming message. Comparing the logged raw value against the time shown on screen confirmed the theory exactly:

```
12:23:16 (raw timestamp from SLT API)
+ 05:30 (incorrectly added offset)
-----
17:53:16 (what the UI displayed)
```

Server clock and Laravel's `APP_TIMEZONE` config were both verified correct beforehand via `php artisan tinker`, ruling out a system clock or config issue and confirming the bug was purely in the parsing logic.

4. The Fix

One line was changed: when a raw timestamp has no timezone offset, it is now parsed as already being in the application's configured timezone (Asia/Colombo) instead of being assumed to be UTC.

File	<code>app/Services/ChatConversationService.php</code>
Function	<code>extractIsoTimestamp()</code>
Before	<code>Carbon::parse(\$raw, 'UTC')</code>
After	<code>Carbon::parse(\$raw, \$tz)</code>

5. Pseudocode

```

FUNCTION extractIsoTimestamp(message):
    targetTimezone = "Asia/Colombo"
    rawTimestamp = GET timestamp/time/sent_at/created_at FROM message

    IF rawTimestamp IS EMPTY:
        RETURN null

    IF rawTimestamp IS NUMERIC (epoch):
        TREAT as UTC epoch, CONVERT to targetTimezone
        RETURN formatted result

    hasOffset = CHECK IF rawTimestamp ENDS WITH "Z" or "+HH:MM" or "-HH:MM"

    IF hasOffset IS TRUE:
        // Timestamp already tells us its zone - trust it
        parsedTime = PARSE(rawTimestamp)
    ELSE:
        // BEFORE (bug): parsed as UTC, causing +5:30 double-shift
        // AFTER (fix): parsed as already being local Sri Lanka time
        parsedTime = PARSE(rawTimestamp, assumeZone = targetTimezone)

    RETURN CONVERT(parsedTime, targetTimezone) AS ISO-8601 string
END FUNCTION

```

6. Remaining Work

- Test the fix against a real incoming message to confirm the time now displays correctly (e.g. 12:23, not 17:53).
- Old messages already saved in the database still carry the incorrect (+5:30 shifted) timestamps — a one-time cleanup/migration is needed if historical data should be corrected too.
- Remove the temporary debug log line (SLT_TS_DEBUG) once the fix is confirmed working in production.